

Phu Vinh Nguyen

Boston, MA | (727) 383 6232 | nphuey@gmail.com | Website: phu-ngu.github.io | github.com/phu-ngu

EDUCATION

Northeastern University	Boston, MA
Certificate in Bioinformatics, concentration in programming	May 2025
Relevant Coursework: Python for Bioinformatics, Genomics in Bioinformatics, Algorithms, R programming	

Uniwersytet Jagielloński	Krakow, PL
Master's in Molecular Biotechnology	Jul 2021

University of California, Irvine	Irvine, CA
Bachelor of Science in Pharmaceutical Sciences	Sept 2019

WORK EXPERIENCE

Lab Automation Engineer Developer , Sanofi Pasteur	Jun 2025 – Present
<i>Automation & Data Engineering Team</i>	

- Developed a form help request desktop app using React and Electron to standardize researcher form submissions and automate internal ticket creation via SMTP email integration.
- Designed, tested, and deployed methods for liquid handler robots to automate lab workflows.

Automation Engineer , Foundation Medicine Inc	Nov 2024 – Apr 2025
<i>Assay Development, Automation Engineering Team</i>	

- Developed a Python GUI app using CustomTkinter to provide a user-friendly interface for backend automation scripts, enabling non-technical users to perform tasks independently.
- Built a React troubleshooting app for Hamilton robot devices to display key metadata in real time, enabling teams to accelerate debugging and improve workflow reliability.
- Created barcode scanning systems and ETL workflows to integrate sample data into Sapio LIMS, automating sample tracking and reducing manual entry errors.

Lab Automation Engineer , Sanofi Pasteur	Oct 2022 – Nov 2024
<i>Automation Team</i>	

- Revised validation protocols and SOPs; trained end-users on automated systems across R&D teams.
- Optimized high-throughput Hamilton methods, reducing assay processing times by up to 30% in DNA extraction and fluorescence-based workflows.
- Delivered biweekly presentations to stakeholders on automation performance, challenges, and improvements supported by technical data.

Senior Research Associate , Dermibiont Inc.	Oct 2021 – Sept 2022
<i>Clinical Diagnostic Team</i>	

- Built data analysis pipelines for Oxford Nanopore Technology (ONT) using open-source bioinformatics tools, including Guppy basecalling and quality control filtering.
- Collaborated with external bioinformatics teams to troubleshoot and optimize Dockerized QC pipelines.
- Automated DNA extraction on Opentrons and visualized results using Python and R.

PROJECTS

Personal Website – phu-ngu.github.io

- Developed a React portfolio website to showcase professional experience, technical projects, and programming skills.

SKILLS

Languages: Python, JavaScript, SQL, HTML, CSS, Bash

Frameworks and Libraries: React.js, Express.js CustomTkinter, Electron

Technical Tools: Git, Docker, Jira, LIMS (user level), HPC environments